

# Database Migration from EC2 (MySql) to RDS for MySQL

## Project Overview:

Project name: MySQL Database Migration from EC2 to RDS  
Source DB: MySQL on EC2  
Target DB: Amazon RDS for MySQL  
Migration Tool: AWS Database Migration Service (DMS)  
Migration Type: Homogeneous Migration  
Migration Approach: Full Load + Change Data Capture (CDC)  
Downtime Objective: Minimal Downtime  
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### 1. Executive Summary:

This document describes the migration strategy for transferring the existing MySQL database hosted on an Amazon EC2 instance to Amazon RDS for MySQL using AWS Database Migration Service (DMS).

The migration will utilize a Full Load followed by Change Data Capture (CDC) to ensure near-zero downtime and data consistency during cutover.

### 2. Objectives:

#### a. Business Objectives:

- Improve database availability and reliability.
- Reduce operational overhead.
- Utilize managed database services.
- Enhance backup and recovery capabilities.
- Improve scalability and monitoring.

#### b. Technical Objectives:

- Migrate all database objects and data.
- Minimize downtime.
- Maintain data integrity.
- Ensure successful application connectivity after migration.

### 3. Current and Target Environment:

| Components | Source System | Target System |
|------------|---------------|---------------|
|------------|---------------|---------------|

|                         |                       |                        |
|-------------------------|-----------------------|------------------------|
| Database Engine         | MySQL                 | MySQL                  |
| Hosting Platform        | Amazon EC2            | Amazon Managed (RDS)   |
| Database Version        | MySQL 8.x             | Compatible with Source |
| Operating System        | Linux                 |                        |
| Database Size           | TBD                   |                        |
| Number of Tables        | TBD                   |                        |
| Instance Type           |                       | db.t3.medium           |
| Storage Type            |                       | GP3                    |
| Backup Retention        |                       | 7 Days                 |
| Multi-AZ                |                       | Yes                    |
| Host (Connectivity)     | Source EC2 Private IP | RDS Endpoint           |
| Port (Connectivity)     | 3306                  | 3306                   |
| Protocol (Connectivity) | TCP                   | TCP                    |

#### 4. Migration Architecture:

MySQL (EC2) -> AWS DMS Replication Instance -> Amazon RDS MySQL

Migration Flow:

1. Full Data Load
2. Continuous CDC Replication
3. Data Validation
4. Application Cutover